

$$\frac{\partial \mathcal{L}}{\partial \omega_{12}} = \frac{\partial \mathcal{L}}{\partial y^1} \frac{\partial y^1}{\partial o_2} \frac{\partial o_2}{\partial \omega_{12}} + \frac{\partial \mathcal{L}}{\partial y^2} \frac{\partial y^2}{\partial o_2} \frac{\partial o_2}{\partial o_1} \frac{\partial o_1}{\partial \omega_{12}}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial \omega_{12}} = \sum_{j=1}^n \frac{\partial \mathcal{L}}{\partial y^j} \frac{\partial y^j}{\partial o_2} \frac{\partial o_2}{\partial \omega_{12}}}$$